Liang–1997 SHE–BTE: Multi-Run Reproducibility Study of the Drain-Refined G_{ii} Trio

Frozen solver tag `solver-frozen-2026-09-26` (git `930dd86`)

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Purpose

Quantify what, in the certified Liang-1997 reproduction, is *reproducible* across independent re-runs of the frozen solver, and what is not. No code, tolerance, or mesh was changed. The certified reference is the 2026-09-24 cold run (pure-Python kernel): $G_{ii} = 3.075619/2.136180/2.387429 \times 10^{25}$ at $71 \times 58/100 \times 70/141 \times 85$; non-monotone, $S_{\mathrm{rev}} = 737$, Richardson suppressed.

Runs compared

run	machine	kernel	conditions
Sep-24 baseline	Windows, 24 GB	pure-Python	certified reference
Windows interrupted	Windows, 24 GB	pure-Python	∼4 OOM resumes
Windows clean	Windows, 24 GB (64 GB pagefile)	pure-Python	uninterrupted
Linux (numba)	Linux, 31 GiB, 24 thr	**numba JIT**	uninterrupted
Linux (pure-Python)	Linux, 31 GiB	pure-Python	*pending* (`SHE2D_DISABLE_NUMBA=1`)

Per-node converged G_{ii} ($\times 10^{25}$)

node	Sep-24	Win. interrupted	Win. clean	Linux (numba)
71×58	3.075619	3.075619	3.075619[†]	2.9847
100×70	2.136180	2.142249	2.087561	∼1.80 (limit-cycle, uncert.)
141×85	2.387429	2.375921	2.399434	—
max rel. dev. vs Sep-24	—	+0.28%	−2.28%	−2.9% (71×58)

[†] Win. clean 71×58 reused the earlier clean-run checkpoint; an independent clean pure-Python derivation reproduced 3.075619×10^{25} to 1.6×10^{-7} (bit-exact).

Trio verdict — reproduces across all runs

run	sequence signs	monotone?	S_{rev}
Sep-24	$(-,+)$	no	737
Windows interrupted	$(-,+)$	no	1018
Windows clean	$(-,+)$	no	2415

Every run yields a significantly non-monotone drain-refined sequence ($S_{\text{rev}} \gg 1$): no real convergence order exists, so Richardson extrapolation is **suppressed** and **there is no justified continuum** G_{ii}. This qualitative verdict is robust to environment, kernel, and interruption.

Why absolute G_{ii} is not reproducible (∼2–3%)

Two mechanisms, both amplified by G_{ii} being a near-null-space, hotspot-sensitive moment:

(1) Plateau-gate false-pass on a transient flat spot. The gate certifies when the last-4 cumulative-iteration spread of $G_{ii,\max}$ falls below 3×10^{-4}. A locally flat window *mid-transient* can trip it early. The Windows *clean, pure-Python* 100×70 did exactly this:

	chunk 3	chunk 4	...	baseline chunk 7
cum iters	240	300		480
G_{ii} ($\times 10^{25}$)	1.949	**2.0876**		2.1362
last-4 spread	1.1×10^{-2}	1.2×10^{-4} **PASS**		1.9×10^{-4} PASS

G_{ii} had just risen +7% (chunk 3→4) — clearly still climbing toward ∼2.136 — yet the last-4 window was momentarily flat, so the gate certified 2.0876×10^{25} (−2.3%) at 300 iters instead of 480. This affects *clean, same-kernel* runs; it is not a memory/OOM/interruption artifact.

(2) numba vs. pure-Python kernel divergence. The Linux box ran the numba JIT assembly kernel (`SHE2D_DISABLE_NUMBA` unset). Though numba "changes only how the same discrete equations are assembled" (FP rounding/order), the 71×58 trajectory diverged from the reference *at chunk 0* (iters 1–60: 2.6433 vs reference 2.5571×10^{25}) and converged to 2.9847×10^{25} (−2.9%). Its 100×70 entered a hotspot limit-cycle (∼1.80×10^{25}, spread stuck at 2–4 $\times 10^{-3}$) that the pure-Python kernel did *not* exhibit. Thus $\sim 10^{-15}$ kernel rounding differences amplify to ∼3% in the converged G_{ii}.

Conclusion

- **Reproducible:** the non-monotone trio ⇒ Richardson suppressed ⇒ no justified continuum G_{ii}. Confirmed in every run; the YELLOW / non-asymptotic verdict stands and is *strengthened*.
- **Not reproducible:** the absolute G_{ii} of the two larger nodes, to better than ∼2–3%, owing to gate false-passes, kernel differences, and path/warm-start sensitivity of a near-null-space quantity. A clean pure-Python 71×58 *is* bit-reproducible, so the solver itself is deterministic per kernel; the irreproducibility is in where/whether the ill-conditioned plateau is certified.

- **Recommended gate-hardening (future, frozen-ref change — not applied):** require flatness sustained over multiple chunks, or a small chunk-to-chunk ΔG_{ii}, so the gate cannot certify mid-transient. This would make absolute G_{ii} reproducible without affecting the (already robust) qualitative verdict.
- **Pending:** Linux pure-Python run (`SHE2D_DISABLE_NUMBA=1`) — the apples-to-apples clean reproduction; this report will be finalized when it lands.